MIL-I-47100(MI)
24 May 1974
SUPERSEDING
MIS-13919
4 June 1966

MILITARY SPECIFICATION

INSULATION SLEEVING, FLEXIBLE, BRAIDED, 200 DEGREES CELSIUS (HIGH TEMPERATURE POLYAMIDE)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high temperature polyamide (nylon) braided flexible sleeving. The sleeving is furnished over a 0.020-inch metal core.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

Military

MIL-I-3190	Insulation Sleeving, Electrical, Flexible, Treated

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 5970

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MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample of sleeving shall meet the requirements of this specification.

3.2 Materials.

3.2.1 Composition and construction.

3.2.1.1 Composition. The sleeving shall be fabricated from high temperature polyamide continuous filament yarn.

3.2.1.2 Construction. The yarn shall be 200 denier, 100 filament, produced with a zero twist. The yarn should be woven over a removable core constructed from size 24 AWG solid diamond drawn bright annealed stainless steel or diamond drawn monel wire. The braid shall be 16 carriers with one end per carrier in a 2 over and 2 under pattern, 45 to 55 picks per inch.

3.2.1.3 Size. The finished diameter of the sleeving shall be 0.040 inch maximum.

3.2.1.4 Color. The color of the sleeving shall be natural off-white.

3.3 Performance characteristics. (Not applicable)

3.4 Environmental conditions.

3.4.1 Burning rate. The sleeving shall be self-extinguishing, and after vertical orientation shall not burn more than 1/4 inch.

3.4.2 Heat aging. The braid shall not crack, fracture, or show brittleness after the heat aging test specified herein.

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3.5 Workmanship. The workmanship shall be such as to ensure a product which is uniform and in conformance with this specification. The sleeving shall be free from defects and imperfections that would render the material unsuitable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. When specified in the contractual document (see 6.2), a sample consisting of 25 feet of sleeving, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein. If the preproduction sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table I, shall be conducted in accordance with the corresponding test paragraphs.

Table I
Preproduction Inspection

Characteristic	Requirement source	Test paragraph
Composition and construction	3.2.1	4.5.3.1
Burning rate	3.4.1	4.5.3.2
Heat aging	3.4.2	4.5.3.3
Workmanship	3.5	4.5.4

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4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 4.0 percent defective.

4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1, to determine compliance with the characteristics specified in Table II, shall be conducted in accordance with the corresponding test and inspection paragraphs.

Table II.

Acceptance Inspection

Characteristic	Requirement source	Test and inspection paragraph
Composition and construction	3.2.1	4.5.3.1
Burning rate	3.4.1	4.5.3.2
Workmanship	3.5	4.5.4
Packaging, packing and marking	5	4.6

4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified

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in the applicable test methods and procedures paragraphs.

4.5 Test methods and procedures.

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature, 24 degrees plus or minus 9 degrees Celsius (C) (75 degrees plus or minus 16 degrees Fahrenheit (F)).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.5.2 Test sequence. Testing within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Tests.

4.5.3.1 Composition and construction. Composition shall be certified by the supplier and construction shall be examined visually to establish compliance to 3.2.1.

4.5.3.2 Burning rate. The core of the sleeving shall be removed. The sleeving shall then be subjected to the burning rate test in MIL-I-3190 and shall conform to 3.4.1.

4.5.3.3 Heat aging. The core of the sleeving shall be removed. The sleeving shall then be subjected to the heat aging test in accordance with MIL-I-3190 except that the temperature shall be 491 degrees plus or minus 9 degrees F (225 degrees plus or minus 5 degrees C) and the duration shall be 24 hours. The sleeving shall conform to 3.4.2.

4.5.4 Workmanship. The sleeving shall be visually examined to assure conformance to the workmanship requirement specified in 3.5.

4.6 Packaging, packing and marking. The inspector shall ascertain that the packaging, packing and marking of the material conforms to this specification.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be level A, B, or C as specified (see 6.2), in accordance with Section 5 of MIL-I-3190.

5.2 Marking. Marking shall be in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following:

a. Manufacturer's name and address.

b. Manufacturer's designation.

c. Lot number.

d. Number of this specification.

e. Date of manufacture.

f. Contract or purchase order number.

6. NOTES

6.1 Intended use. The sleeving is intended for use as mechanical protection for wire which is subjected to elevated temperatures during short duration exposure to rocket engine exhausts.

6.2 Ordering data. Procurement documents should specify but not be limited to, the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Requirement for certification of composition (see 4.5.3.1).

d. Special marking, if required.

e. Selection of applicable level of preservation, packaging, and packing.

6.3 Supersession data. This specification includes the requirements of MIS-13919, dated 4 June 1966.

Custodian:
Army-MI

Preparing Activity:
Army-MI

Project No. 5970-A434

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-47100	Insulation Sleeving, Flexible, Braided, 200 Degrees Celsius

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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